Spanish–Asturian Neural Machine Translation

via Transfer Learning from a MarianMT Checkpoint

WMT24 Low-Resource MT Shared Task — Extended Abstract

Darragh Kerins
University of the Basque Country (UPV/EHU)
drussell001@ikasle.ehu.eus

Abstract

We present a bilingual Spanish–Asturian MT system built by fine-tuning `Helsinki-NLP/opus-mt-es-ca` on the `projecte-aina/ES-AST_Parallel_Corpus`. Evaluated on FLORES+ `devtest`, the fine-tuned model achieves SacreBLEU **16.52** versus a zero-shot baseline of 3.24 (+13.28), trained for one epoch on a single NVIDIA RTX 4060. We justify all design choices and identify concrete directions for improvement.

1. Approach and Justification

The WMT24 constrained track [1] limits pre-trained models to $\leq$1B parameters and restricts data to official shared task resources. Fine-tuning a compact pre-trained model was the most practical approach within these constraints.

Model choice. I selected `opus-mt-es-ca` [2] (~74M params) over larger multilingual models such as NLLB-200 for two reasons: (1) Catalan is structurally close to Asturian — both are Western Romance languages with significant lexical overlap — making the weights a more informed initialisation; (2) the shared SentencePiece vocabulary required no tokeniser modifications.

Training decisions. Adafactor [5] was preferred over AdamW to reduce optimizer state memory (~40% lower VRAM), enabling a larger effective batch on the RTX 4060. Gradient checkpointing was enabled for the same reason. One epoch on a 100k subsample was chosen as a reproducible proof-of-concept; the script scales to the full corpus via `-full`.

2. Data

Training. `projecte-aina/ES-AST_Parallel_Corpus` [3] provides ~704k Spanish–Asturian pairs. A 100k subsample was used (90k/10k train-val, seed 42), truncated to 128 subword tokens.

Evaluation. FLORES+ `devtest` [4]: 1,012 held-out `spa_Latn` $\rightarrow$ `ast_Latn` pairs, unseen during training.

3. Results

Table 1: SacreBLEU [6] on FLORES+ `devtest`.

System	BLEU
Baseline (`opus-mt-es-ca`, zero-shot)	3.24
Fine-tuned (1 epoch, 100k pairs)	**16.52**

The +13.28 gain confirms the transfer hypothesis: the Catalan checkpoint adapts quickly to Asturian even with minimal training. The baseline scores low because it produces fluent Catalan, not Asturian. After fine-tuning, the model acquires characteristic Asturian patterns: *-u* nominal endings (*departamentu*), clitics (*comunica-y*), contractions (*l'Asociación*), and the negation particle *nun*.

4. Critical Analysis and Future Work

Limitations. This is a bilingual system and is best understood as a strong baseline. Only one epoch on 14% of available data was used, leaving clear headroom. The tokeniser was not adapted for Asturian, so infrequent surface forms are likely under-represented. FLORES+ Asturian references are community-contributed and may be inconsistent, which lowers the effective BLEU ceiling.

Potential improvements. (1) Training on the full 704k pairs for 3+ epochs is the lowest-effort gain. (2) A **multilingual setup** adding related Romance languages (Galician, Portuguese) could improve low-resource generalisation via cross-lingual transfer. (3) **PEFT** (e.g. LoRA) would allow vocabulary-adapted fine-tuning within the same VRAM budget. (4) Adding Asturian-specific tokens to the tokeniser, initialised from phonologically similar Catalan embeddings, would address the subword coverage gap directly.

Submitted Files

- `finetune_es_ast.py` — local GPU training script with VRAM-aware batch tuning
- `Spanish_Asturian_MT.ipynb` — Colab notebook: baseline vs. fine-tuned eval on FLORES+
- `flores_devtest_hypotheses_ast.txt` — 1,012 system hypotheses, one per line

References

[1] WMT24 Organizers. 2024. *Findings of the WMT24 Shared Task on Low-Resource MT*. *Proc. WMT*. https://aclanthology.org/2024.wmt-1.57/

[2] J. Tiedemann and S. Thottingal. 2020. *OPUS-MT*. *Proc. EAMT*.

[3] Projecte AINA. 2023. *ES-AST Parallel Corpus*. https://hf.co/datasets/projecte-aina/ES-AST_Parallel_Corpus

[4] NLLB Team. 2022. *No Language Left Behind*. arXiv:2207.04672.

[5] N. Shazeer and M. Stern. 2018. *Adafactor*. *Proc. ICML*.

[6] M. Post. 2018. *A Call for Clarity in Reporting BLEU Scores*. *Proc. WMT*.